ELECTROCHEMISTRY UNPLUGGED: A VISUAL GUIDE

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ELECTROCHEMICAL CELL (VOLTAIC/GALVANIC)

AT ANODE: Oxidation ($Zn \rightarrow Zn^{2+} + 2e^-$)

AT CATHODE: Reduction ($Cu^{2+} + 2e^- \rightarrow Cu$)

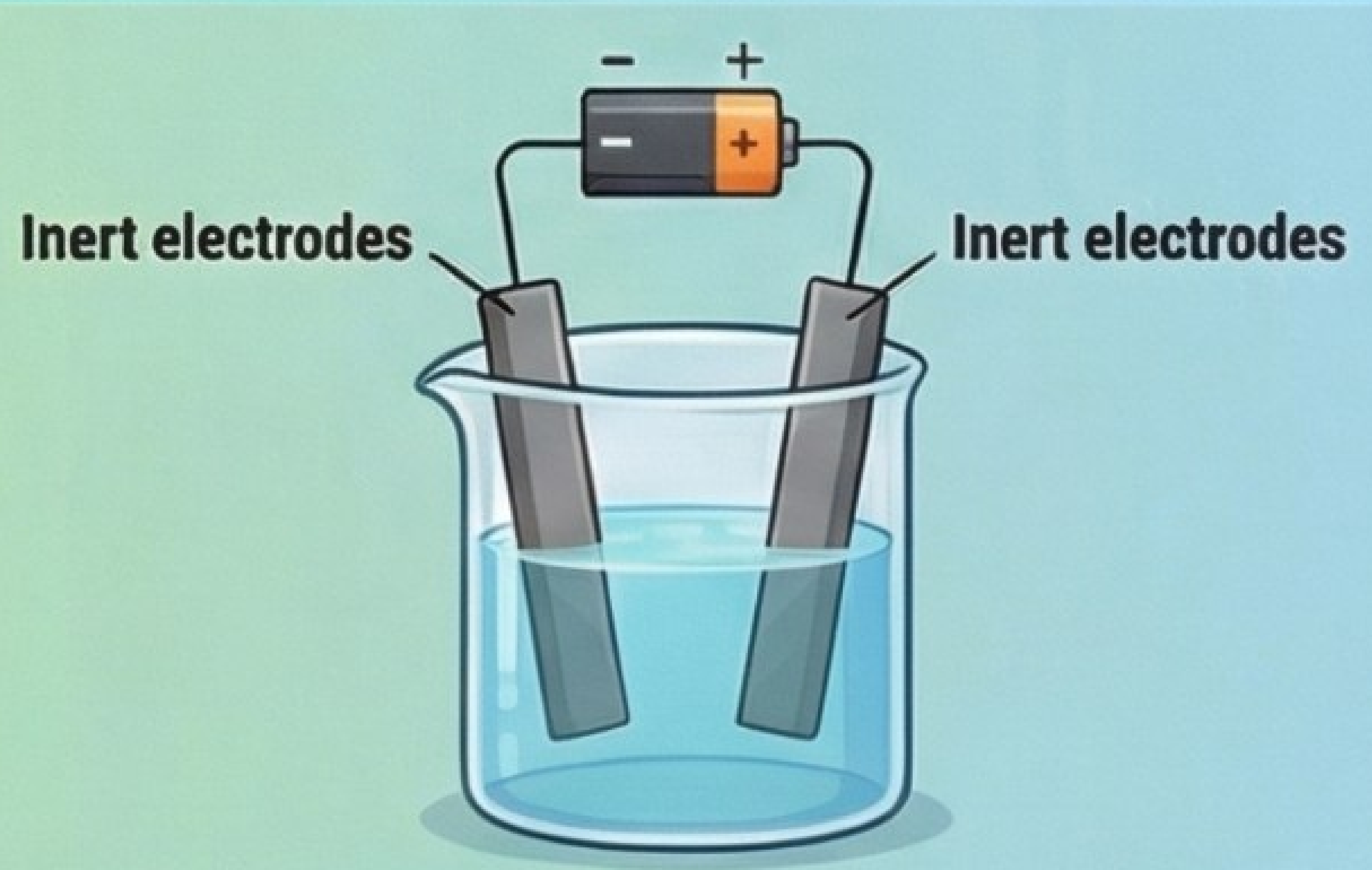
Cell Representation: $Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

Current flows from the Copper cathode to the Zinc anode, opposite to the electron flow.

ELECTROCHEMISTRY:

The bridge between chemical and electrical energy. Involves **Chemical ↔ Electrical** energy conversions.

ELECTROLYTIC CELL



TWO SIDES OF THE SAME COIN

	Electrochemical Cell	Electrolytic Cell
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Anode Charge	Negative (−)	Positive (+)
Cathode Charge	Positive (+)	Negative (−)
Process	Spontaneous	Non-spontaneous

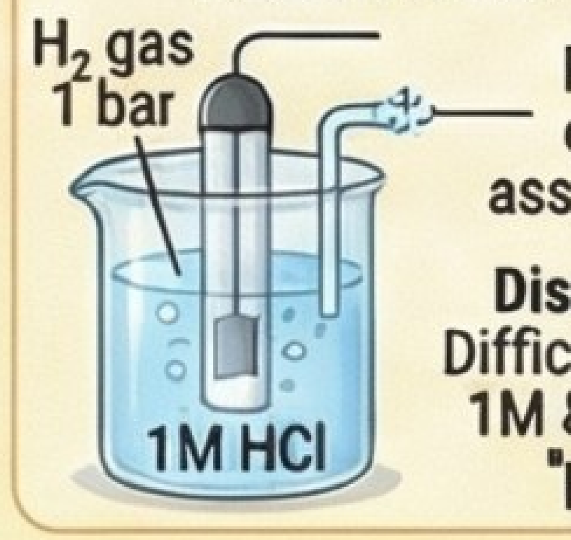
DRIVING FORCE: ELECTRODE & CELL POTENTIAL

Electrode Potential (E)
Potential difference at electrode-electrolyte interface due to charge separation.

Standard Electrode Potential (E°)
Measured at 1M concentration under standard conditions.

$$E^{\circ}_{cell} = E^{\circ}_{(right)} - E^{\circ}_{(left)} = E^{\circ}_{(cathode)} - E^{\circ}_{(anode)}$$

THE BENCHMARK: STANDARD HYDROGEN ELECTRODE (SHE)



Reference electrode, assigned 0.00 V

Disadvantages: Difficult to maintain 1M & 1 har; easily "poisoned".

BEYOND THE STANDARD: NERNST EQUATION & GIBBS ENERGY

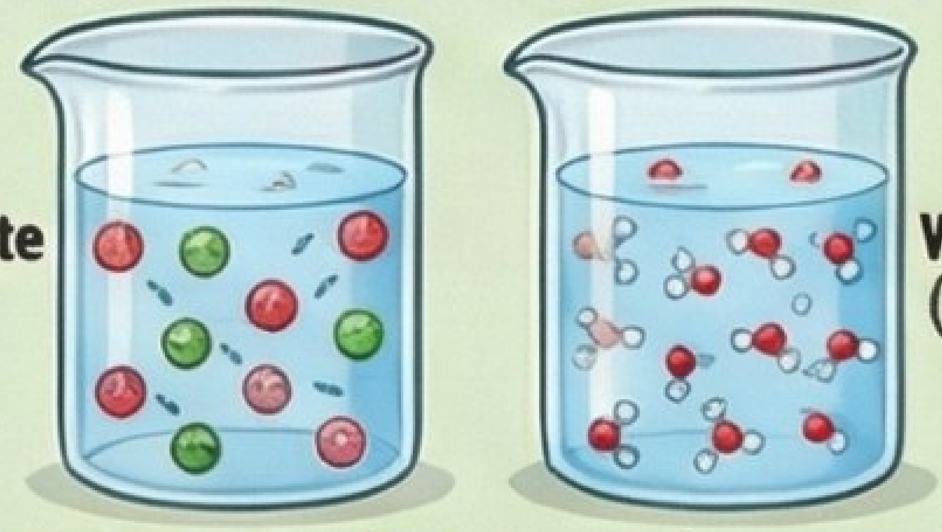
Nernst Equation (Non-standard conditions):

$$E_{cell} = E^{\circ}_{call} - (0.0591/n) * \log(Q_c)$$

(where n = moles of electrons and Q_c is the reaction quotient: [products]/[reactants])

Gibbs Free Energy (ΔG): Relates cell potential to spontaneity.
Standard: $\Delta C^{\circ} = -nFE^{\circ}_{cell}$; **Non-Standard:** $\Delta G = -nFE_{cell}$. Negative ΔG indicates spontansity.

HOW SOLUTIONS CONDUCT ELECTRICITY

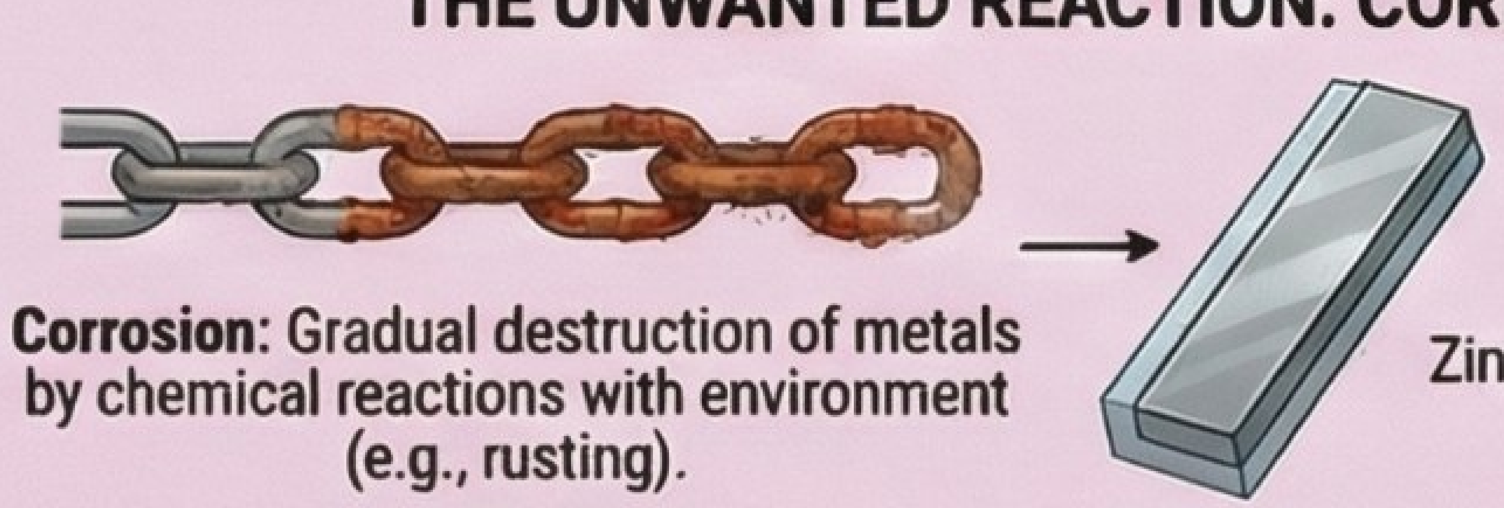


Strong Electrolyte (e.g., NaCl)

Weak Electrolyte (e.g., CH_3COOH)

EFFECT OF DILUTION: Conductivity (e) decreases (fewer ions/volume), **Molar Conductivity (Λ_m) increases** (total volume increases). For weak electrolytes, Λ_m rises sharply due to increased dissociation.

THE UNWANTED REACTION: CORROSION



Corrosion: Gradual destruction of metals by chemical reactions with environment (e.g., rusting).

A SMART PREVENTION: GALVANIZATION

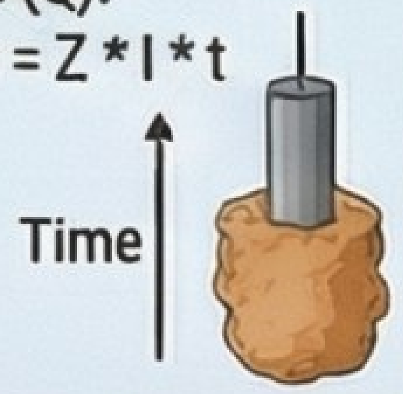
Iron coated with a more reactive metal (Zinc). Zinc corrodes firat. "sacrificing" Rself to protect iron (Sacrificial method).

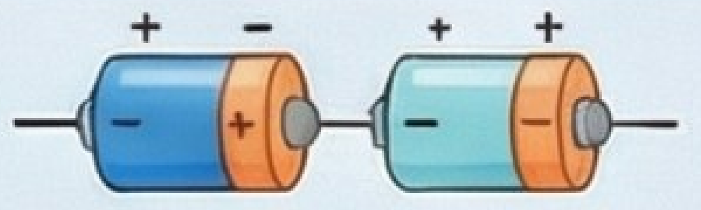
THE LAWS OF ELECTROLYSIS (FARADAY'S LAWS)

Faraday's First Law

Mass (w) of substance deposited ∝ Total electric charge (Q).

Formula: $w = Z * I * t$





Faraday's Second Law

For same charge through different electrolytes in series, mass deposited ∝ chemical equivalent weights (E).

Formula: $\frac{w_1}{w_2} = \frac{E_1}{E_2}$